

'animate()' is used for this purpose.

Dice.h. It is used to declare the subroutines used in the project. This file also defines the variables and constants in relationship with I/O ports that have been used to interface MAX7219 with the microcontroller. It also defines the variables and constants associated with memory locations of MAX7219.

MAX7219.c. It is used to control the flow of data to MAX7219 and subsequently to the LED matrix. The main subroutine used for generating serial clock to control the flow of data and activating/enabling MAX7219 is achieved by 'send_Byte_7219(char data)' subroutine. The clock is generated by controlling 'CTRL_CLK I/O' interface line (with 'cbi' and 'sbi' instructions). Data is transferred to MAX7219 in serial format through controlling line 'CTRL_DATA I/O'.

All 16-bit information has to be sent to MAX7219 for proper operation. First, the most significant byte (MSB) is sent, which contains address of the desired purpose. Then the actual data is sent. This is achieved by sending two 8-bit individual data to MAX7219 with 'write_7219 (char reg_number, char data)' subroutine.

With these two subroutines, the operation of MAX7219 can be controlled to display the desired data on the LED matrix. The other subroutines used in the Max7219.c file are listed in Table III.

'disp_Char(char character)' subroutine is used to show the information on the LED matrix display. The character

PARTS LIST

Semiconductors:

IC1	- ATmega8 AVR microcontroller
IC2	- MAX7219 LED display driver
IC3	- 7805, 5V regulator
D1-D4	- 1N4007 rectifier diode
LED1	- 5mm LED
DIS1	- 8×8 column-cathode dot-matrix display

Resistors (all 1/4-watt, ±5% carbon):

R1	- 470-ohm
R2	- 220-ohm
R3, R4	- 10-kilo-ohm
R5	- 12-kilo-ohm

Capacitors:

C1	- 1000µF, 25V electrolytic
C2	- 0.1µF ceramic disk
C3, C4	- 33pF ceramic disk
C5	- 10µF, 16V electrolytic

Miscellaneous:

X1	- 230V AC primary to 9V, 500mA secondary transformer
S1, S2	- Push-to-on tactile switch
X _{TAL}	- 4MHz crystal

or any other information that has to be shown on the LED display is first converted into an LED map, i.e., which LED will be on or off (digit-wise). The subroutine reads the 5x7 font data that has been declared for each and every character (and desired information) in 'font5x7.h' ('font5x7' array is used for the purpose). Each character's font data is given in five columns (selected digit) and seven rows (segment data). This data is basically read as one column (five columns in all) at a time into an array named 'matrix[]'.

First, the font data is read and then with available 'matrix[]' information, 'display_matrix()' subroutine is used to write the data into MAX7219's

digit memory location. 'Matrix[]' data controls the segment I/O lines of MAX7219 to light the corresponding LED of the selected digit of the display. Once the display data is transferred, the digits are scanned automatically one after the other to show one digit for the set time period.

Operation

1. Once the hardware is powered, it immediately enters the animated screen where two patterns are displayed one after the other in quick succession.

2. At this point of time, the user can press the key. The display pattern freezes on the display and remains in this condition until a key is pressed.

3. As soon as the key is released, the dice-rolled number is displayed on the screen for a set time (as programmed in the source code).

4. After a preset time period, the program resumes to point-1 operation described above.

5. If the pushbutton is pressed again while the dice-roll number is being displayed and released, then also a new dice-rolled number is displayed on the screen. This kind of operation also brings in an element of randomness.

Fig. 6 shows the project in various stages of operation.

EFY note. The source codes of this article are available on www.efymag.com website.

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